



Using Machine Learning to downscale climate scenarios

A robust statistical downscaling framework for regional adaptation

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The Resolution Gap

- **The problem:** Global Climate Models typically resolve temperature at roughly **100-150 km**.
 - This is enough for synoptic circulation and large-scale forcing.
 - It is not enough for mountain gradients, coastlines, valleys, and local exposure.
- **The user need:** adaptation studies usually need information closer to **10 km or finer**.
 - agriculture, health, hydrology, infrastructure, and demonstrator services all need local structure
- **What is missing in coarse pixels:**
 - topography-induced gradients
 - land-sea contrasts
 - regional spatial organization
- **This is the downscaling problem:** reconstruct fine-scale structure that is consistent with the large-scale climate driver.



Starting Point For This Short Course

- **We should not assume the same background for everyone**
 - Some of you know reanalyses and observations very well
 - Others know impact workflows or bias correction better than ML
- **So we start from shared climate notions**
- **The new part is not the climate problem itself.**
 - ML changes how we learn the coarse-to-fine mapping.
 - It does not remove the need for domain knowledge or evaluation discipline.
- **Take-home framing:** AI downscaling is still a climate workflow before it is an ML workflow.

The "Perfect Model" Strategy

- **Objective of the method:** learn a transfer from coarse climate information to local daily structure, then apply it to climate scenarios.
- **Immediate difficulty:** we do not have perfect daily GCM-observation supervision because the weather chronology does not match.
- **Why not train directly on GCM vs observations?**
 - daily weather chronologies do not match between historical GCM and observations
 - we cannot supervise day 2003-07-12 in a GCM against observed day 2003-07-12 and call that a true pair

The "Perfect Model" Strategy

- **The workaround:** create a controlled laboratory using **ERA5 reanalysis**.
 - Coarsen ERA5 to a GCM-like scale.
 - Keep E-OBS at the target high resolution.
 - Learn the mapping from coarse ERA5 predictors to high-resolution E-OBS.
- **Why this helps:** the chronology is aligned day by day, so the network sees physically consistent predictor-target pairs.
- **Inference idea:** once trained, reuse the learned mapping on climate-model inputs that have been brought closer to ERA5 statistics.



Where ML Actually Sits

- **Inputs:** coarse predictors, static fields, and a carefully defined target grid.
- **Model task:** learn the mapping from large-scale information to local spatial structure.
- **Output:** a high-resolution field, not a replacement for climate science judgment.
- **Practical point:** the model learns the prepared dataset contract, not some abstract "truth".

Where ML Actually Sits

Temperature Downscaling Workflow

A workflow-aligned view of preprocessing, bias correction, ML inference, and evaluation for the exp5 case study

TRAINING LABORATORY

Phase 1 · ERA5 → E-OBS pairs

- Coarsen ERA5 to a GCM-like scale
- Add topography as a static predictor
- Create daily supervised pairs on the E-OBS grid

Input / target setup

- Inputs: elevation + coarse temperature
- Target: daily E-OBS temperature
- Same scaling and layout at inference time

U-Net reference model

- Train U-Net on ERA5 → E-OBS pairs
- Keep one historical checkpoint as reference
- Compare replay and final metrics to archive

PRODUCTION AND VALIDATION PATH

Phase 2 · Bias correction

- GCM historical / future volumes
- Ibis CDF-t cross-calibration
- Separate train / test / future volumes

Phase 3 · Package samples

- Raw GCM test samples
- GCM-BC test samples
- Same layout the model expects

Phase 4 · Inference

- Apply trained U-Net to GCM-BC
- Produce historical downscaled fields
- Then compare with the reference run

Validation stack

- Daily / monthly metrics
- VALUE summary
- PDFs, maps, residuals, parity tests

Message: preprocessing defines the problem, the model learns the prepared inputs and targets, and evaluation must combine metrics, distributions, and maps.



What a Neural Network Is In This Context

- **A neural network is a parameterized function**
 - Inputs go in as arrays on a grid.
 - Outputs come out as another grid of the same or different resolution.
- **Training means adjusting weights from examples**
 - Compare prediction to the target
 - Compute a loss
 - Update millions of weights to reduce that loss



What a Neural Network Is In This Context

- **Important practical point:** the loss definition matters
 - the model learns what the loss rewards
 - scaling, masking, and target definition therefore become scientific choices
- **What the network is not doing:**
 - it is not discovering climate laws from scratch
 - it is not guaranteed to preserve physical behavior unless the data contract and evaluation enforce it
- **So the scientific question is:** what mapping did the data and loss actually teach the network?

Why U-Net Is A Good Fit Here

- **U-Net is a grid-to-grid convolutional neural network**
 - It takes an image-like field in
 - and predicts an image-like field out
- **Encoder path:** compresses the input and captures large-scale context
- **Decoder path:** reconstructs a local high-resolution field
- **Skip connections:** pass spatial detail from early layers to later layers
 - This is important for boundaries, coastlines, and localized structure
- **Why it fits downscaling:** we need both
 - The coarse synoptic context
 - and the precise local placement of gradients and patterns

Why U-Net Is A Good Fit Here

What U-Net Does In This Downscaling Workflow

A U-Net is a grid-to-grid convolutional network: it compresses large-scale context, then reconstructs a local high-resolution field with skip connections.

INPUTS, ENCODER, DECODER, OUTPUT

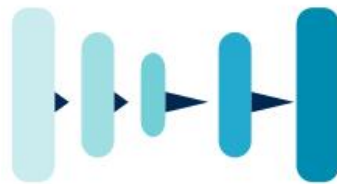
1. Inputs

- Coarse daily temperature field
- Static topography channel
- Normalized using training statistics

What enters the network?

A small stack of gridded predictors.
The network sees arrays, not climate theory, and learns from examples.

2. Encoder → bottleneck → decoder



Skip connections help recover local structure

- Encoder gathers large-scale context
- Bottleneck compresses information
- Decoder rebuilds local structure on the target grid

3. Prediction target

- One daily E-OBS temperature field
- 64×64 France grid
- Optimized with a temperature loss

What is learned?

A mapping from coarse atmospheric state to local spatial temperature structure: relief, coastlines, and regional gradients.

4. Why it helps

- BC adjusts coarse statistics
- U-Net adds local spatial detail
- Both are needed in this workflow

Why not use BC alone?

Bias correction does not invent mountain gradients or coastline structure inside a coarse GCM cell.

Key takeaway: a U-Net combines coarse context and local placement for supervised climate downscaling.

What our Model Actually Sees

- **Inputs**

- Coarse daily temperature predictor
- Static topography as an additional channel

- **Target**

- Daily E-OBS temperature on a **64x64 France grid**

- **Normalization**

- Training and inference must use the same statistics and scaling assumptions
- This is part of the method, not post-processing decoration

- **Core training choices**

- Deterministic U-Net
- Masked MSE-style objective for temperature
- Checkpoint selected from the historical training workflow

Why Use ML In Addition To Bias Correction

- **Bias correction alone is useful but limited.**
 - it adjusts coarse-grid statistics
 - it does not create new local spatial structure inside a coarse GCM cell
- **ML adds a learned spatial transfer.**
 - relief-related gradients
 - coastline contrasts
 - regional spatial organization learned from ERA5 → E-OBS pairs
- **Why BC still matters before ML inference:**
 - the network was trained on ERA5-like coarse predictors
 - BC makes GCM inputs statistically closer to that training world
- **Practical view:** BC reduces domain shift; ML reconstructs local structure.



Why Preprocessing Is Central

- **Not a side task:** regridding, masks, domains, calendars, units, and physical corrections define the learning problem
 - If these change, the predictor family seen by the model changes
 - The model is then asked to solve a different problem than the one it was trained on
- **This is why "same architecture" is not enough**
 - A good network cannot rescue a scientifically inconsistent dataset
- **Short version:** preprocessing is where climate assumptions become ML inputs



A Real Case Study From This Pipeline

- **Observation during implementation:** a preprocessing mismatch created predictor errors of roughly **8-19 K** before the model even ran
- **Concrete cause:** the coarse ERA5 predictor was rebuilt with a different **ERA5 → GCM-like → target-grid** remapping chain than the intended scientific workflow
- **Interpretation**
 - The network was not failing yet
 - The inputs were already wrong
- **Lesson:** regridding choices are not implementation trivia. They can completely change the field the network learns from and predicts from



The "113 K" Bias Lesson

- **The trap:** train directly on absolute Kelvin targets around 280-300 K
- **What happens with MSE**
 - The loss scale becomes very large immediately
 - Gradients become unhelpfully dominated by absolute magnitude
- **Observed failure mode:** RMSE plateau around **113 K**, weights saturate, and training does not learn the intended mapping
- **What we want instead:** bounded normalized targets so the optimization starts in a learnable range
- **Lesson:** target normalization is part of the model contract, not an optional cosmetic trick

Typical Preprocessing Traps

- **Trap 1: Regridding is not neutral**
 - Different remapping paths can yield meaningfully different predictors
- **Trap 2: Masks and coastlines matter**
 - NaN footprints and land-sea treatment can leak wrong values
- **Trap 3: Topography handling matters**
 - Orography and lapse-rate choices strongly affect temperature
 - A coarse GCM cell and a finer reanalysis grid do not represent the same altitude distribution
- **Trap 4: Time alignment matters**
 - "same day" is not enough if clocks or aggregations differ
- **Trap 5: Training and inference must match**
 - The model expects the same preprocessing contract it saw during calibration

ML Objective Traps

- **Trap 1: Optimizing the wrong loss.**
 - what the model is rewarded for may not be what the user needs scientifically
- **Trap 2: Smoothing can improve RMSE while reducing usefulness.**
 - better scores can come with weaker gradients and weaker extremes
- **Trap 3: Sampling design matters.**
 - small or unrepresentative samples can create false confidence
- **Trap 4: Future-signal preservation is not automatic.**
 - a technically strong model can still distort the climate-change signal

Validation And Verification Must Be Multilayered

- **Scalar metrics:** mean bias, RMSE, maxdiff
- **Distribution diagnostics:** PDFs, tails, extremes
- **Spatial diagnostics:** maps, residual maps, mask/coastline behavior
- **Change-signal diagnostics:** historical vs future consistency
 - Even when not all future figures are shown in this short course, this remains a required validation layer
- **Key message:** never certify a workflow from one pleasing summary metric

Why VALUE Matters

- **VALUE protocol** evaluates downscaling as climate information, not just as an ML output
 - Source: the COST VALUE framework for downscaling evaluation
- **What it adds beyond generic ML scores**
 - Bias
 - Variability
 - Temporal behavior
 - Spatial consistency
- **Current historical VALUE-style summary for our temperature case study (U-Net, GCM-BC)**
 - Spatial correlation is very high
 - Spatial RMSE remains non-zero and must be read with maps
 - Quantile and autocorrelation checks help reveal what scalar means hide

Phase 1 - Learning Temperature Downscaling (France Domain)

- **Data sources**
 - ERA5 reanalysis as the predictor laboratory
 - E-OBS as the high-resolution daily target
 - Static **topography** as a persistent explanatory field
- **Target grid:** France-focused **64x64** domain
- **Task:** learn a mapping from coarse-scale ERA5 predictors to high-resolution E-OBS temperature fields
- **Output of Phase 1:** supervised daily (.npz) files training pairs plus normalization statistics
- **Think of this phase as the calibration laboratory**

Phase 2 - Regional Bias Correction (Europe Domain)

- **Data sources**
 - **CNRM-CM6-1** historical and future simulations
 - **ERA5** as the bias-correction reference
- **Scale change:** from local France target space to broader regional Europe driver space
- **Method: Ibicus CDF-t** cross-calibration
- **Volume design**
 - Train historical period: `train_hist = 1980-1999`
 - Historical period: `test_hist = 2000-2014`
 - Future period: `test_future = 2015-2100`
- **Why isolate the volumes:** it avoids mixing historical and future correction logic too early and makes the reanalysis gap explicit
- **This is where classical bias correction enters the workflow**

Phase 3 And 4 - Packaging, Inference, Evaluation

- **Phase 3 data products**
 - Raw GCM test samples
 - GCM-BC test samples
 - Same sample layout and normalization the model expects
- **Phase 4 inference**
 - Use an existing trained checkpoint when the setup matches, **or retrain for a new setup**
 - Predict historical and later future high-resolution fields
- **Phase 4 evaluation products**
 - Daily metrics
 - Monthly metrics
 - VALUE summary
 - Maps, PDFs, plots, tables
- **This is the handoff from scientific preparation to operational execution**

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Validation stack

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- VALUE summary
- PDFs, maps, residuals, parity tests

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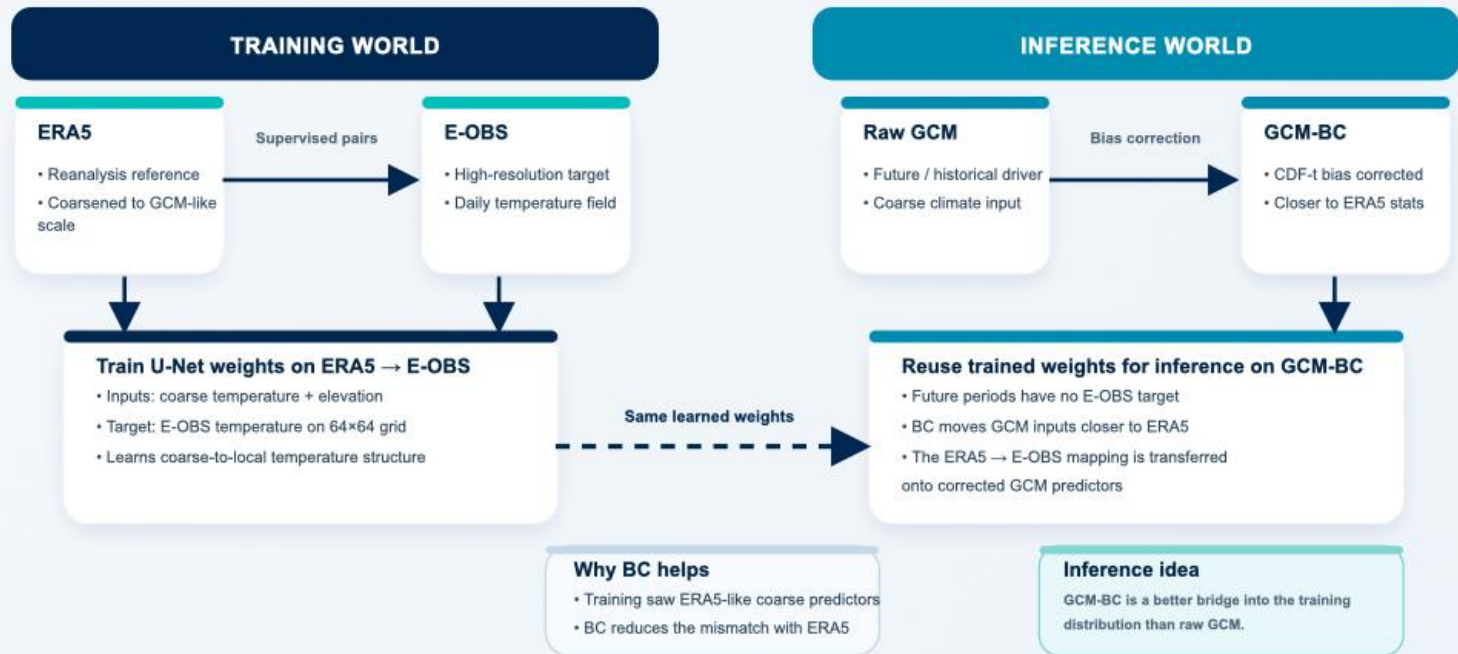
Method Logic - Train On Reanalysis, Transfer To GCM-BC

- **Training world:** ERA5 and E-OBS are aligned day by day, so the network can learn a supervised mapping
- **Inference world:** raw GCM chronology is available, but E-OBS targets are not
- **Therefore**
 - Correct GCM predictors first
 - Bring them closer to ERA5 statistics
 - Reuse the trained ERA5 → E-OBS weights on the corrected GCM inputs
- **Key idea:** BC is not the final answer by itself; it is the bridge that makes the ML transfer more credible

Method Logic - Train On Reanalysis, Transfer to GCM-BC

Method Logic: Train on Reanalysis, Transfer to Bias-Corrected GCM

The learned ERA5 → E-OBS mapping is reused at inference time once GCM statistics are brought closer to the training world.



Key point: train once on ERA5→E-OBS, then apply the same weights to bias-corrected GCM inputs.

Training Is Part Of The Method

- Phase 1 prepares supervised training pairs; it does not train the network by itself
- A **trained checkpoint** can be reused **only when** the preprocessing and data setup remain compatible
- Changing the training reference, predictors, grid, normalization, or domain usually **requires retraining**
- For example using a new reanalysis such as CERRA — the training workflow must be available too
- **A reusable downscaling package therefore needs both**
 - checkpoint-based inference
 - training-capable infrastructure

Weights alone are not the method: the data and preprocessing matter too

VALUE Summary - Useful, But Not The Whole Story

- **VALUE-style summary tells us several things quickly**
 - Temporal behavior is good
 - Spatial correlation is high
 - Quantile and variability diagnostics remain informative
- **But the VALUE table is not enough alone**
 - It does not show spatial texture
 - It does not reveal oversmoothing visually
 - It does not explain where errors are located
- **So VALUE should sit beside maps and distributions, not replace them**

VALUE Summary - Useful, But Not The Whole Story

VALUE-Style Summary For The Historical Temperature Case Study

Case study: exp5 temperature setup, U-Net prediction, bias-corrected GCM input.

SUMMARY TABLE + HOW TO READ IT

Metric	Historical value	How to read it
Mean bias	-0.748 K	Cold overall tendency remains visible.
Std ratio	1.039	Spread is close to observed variability.
q5 / q50 / q95 bias	-1.27 / -0.73 / -0.28 K	Cold tail is the most sensitive part.
Autocorrelation bias	+0.0066	Temporal persistence is very close.
Spatial correlation	0.9933	Large-scale spatial pattern is excellent.
Spatial RMSE	0.817 K	Residual local error still needs maps.

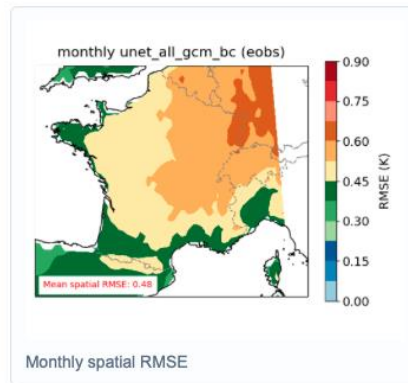
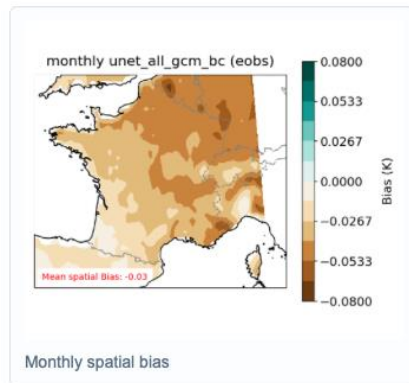
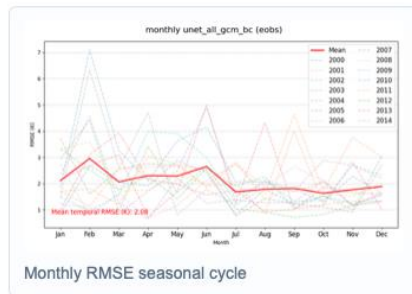
Interpretation

- This is a good operational summary, not the full verdict.
- Very high spatial correlation does not mean perfect local structure.
- Quantile biases show where the tails remain harder to match.
- Maps and PDFs are still needed to detect oversmoothing or texture issues.
- In this project the VALUE-style summary is strongest when paired with:
 - monthly bias / RMSE maps
 - distribution comparison figures
 - archival parity checks

Message: a VALUE table compresses several climate-relevant checks, but it should not be the only thing shown for ML downscaling.

Evaluation Must Be Scientific, Not Cosmetic

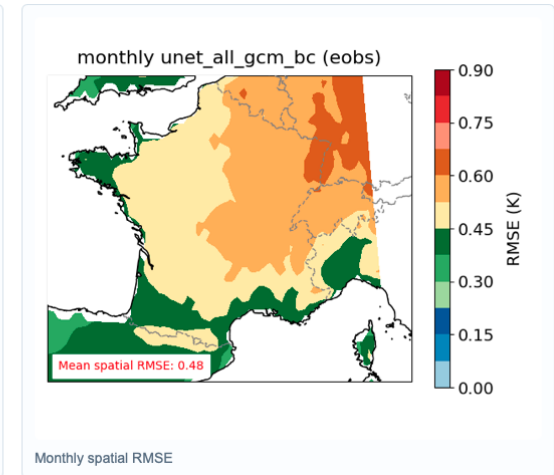
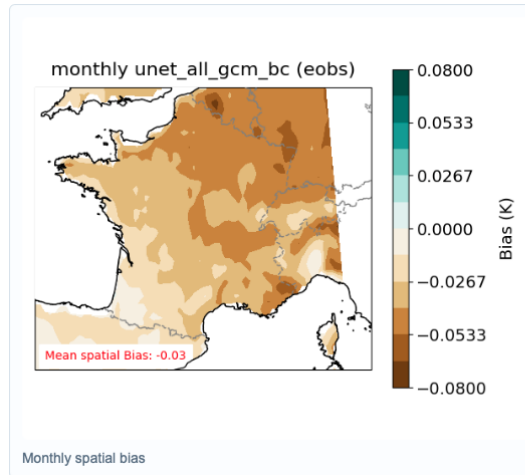
- RMSE is not enough
- We need
 - Maps
 - Distributions / PDFs
 - Regional structure
 - Climate-relevant metrics
- **Key point:** a model can look good globally while being spatially wrong
- **The monthly maps on this slide are the concrete example**



Real Failure Mode - Small Bias, Large Local Error

- **Trap:** claiming success from a small mean bias
- **Missed signal:** local error patterns can remain large and spatially structured
- **Further checks needed**

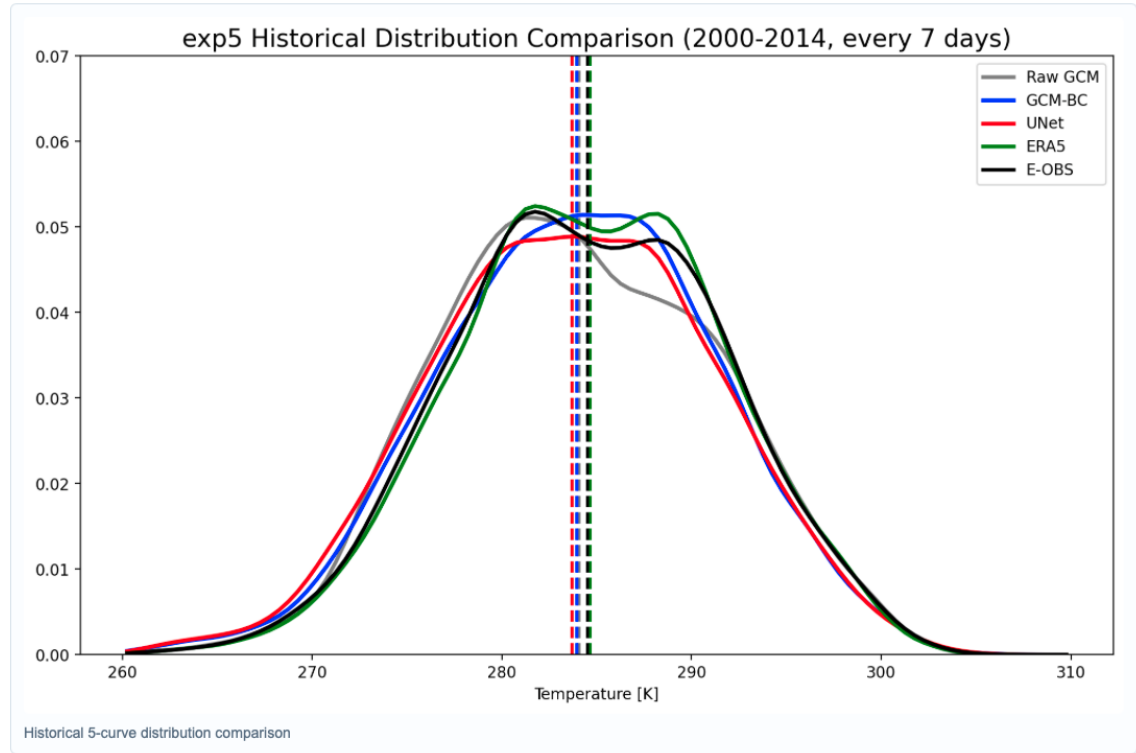
- Monthly spatial bias map
- Monthly spatial RMSE map
- Seasonal curves for context



- **Lesson:** good summary statistics do not guarantee a usable climate field

The 5-Curve PDF Is Powerful - But Partial

- **Why keep the 5-curve PDF**
 - It shows the full story in one view
 - Raw GCM, corrected GCM, ML output, ERA5, and E-OBS
- **Why it is not enough alone**
 - Small visual gaps can still be important
 - It does not say anything about coastline artifacts or oversmoothing
- **Use it as a distribution overview, not as the final verdict**



Distribution Comparison – What BC And UNet Actually Change

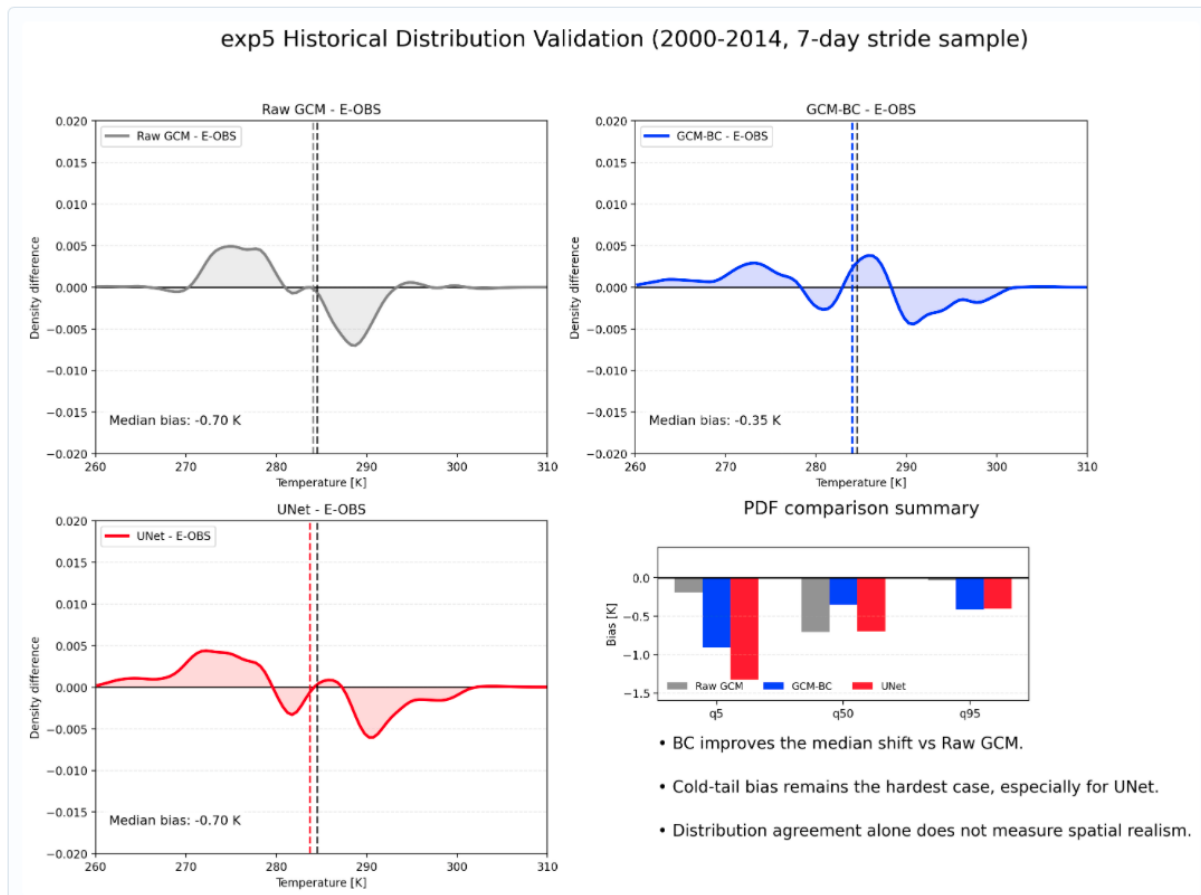
- The point of this figure is not to declare a single winner

- It isolates the three scientifically meaningful comparisons
- Raw GCM vs E-OBS
- GCM-BC vs E-OBS
- UNet vs E-OBS

- Why difference curves

help: they show the magnitude and location of distribution mismatch much more clearly than overlaid PDFs alone

- **Important warning:** distribution agreement does not automatically imply good spatial realism



Computational Architecture - From Laptop To HPC

- **HPC node:** NVIDIA Grace Hopper / ARM-native environment
- **Portable path:** the same workflow logic can be demonstrated locally on short date windows, even if the full production run belongs on HPC
- **Important engineering gain from this work**
 - Restart points
 - Explicit phase boundaries
 - Generic workflow runner
- **Significance:** when climate AI becomes operational, reproducibility and throughput matter as much as model scores

Compute And Scalability – Why Workflow Design Matters

- **Observed wall-clock order of magnitude on our single HPC node for this workflow**
 - Phase 2 Prepare BC reference data: **~5-6 h**
 - Phase 3 Bias-correct GCM fields + packaging: **~1 h**
 - *Optional Training*: about 14–16 s per epoch for our U-Net setup.
30 epochs is **~7–8 minutes**, plus a short test pass.
 - Phase 4 Inference: Historical & Prediction + metrics: **~3 h**
 - Final plotting: **minutes**
- **Operational lesson**: once the workflow is modular and restartable, larger validation campaigns and ensemble-style use become realistic
- **Scientific implication**: if the method is fast enough after training, we can spend more effort on evaluation breadth rather than on one-off heroic runs
- **Retraining is realistic** when the setup changes, instead of forcing users to rely on one frozen checkpoint

Implementation Traps - HPC And Numerics

- **Trap 1: The standardization leak**
 - One code path opened data directly instead of going through the standard cleaning and normalization steps
 - Result: a catastrophic -273 K style failure
- **Trap 2: The noon / midnight shift**
 - GCM timestamps and ERA5 timestamps did not align naively
 - Solved by explicit year-month-day matching rather than hand-waving "same day"
- **Trap 3: Parallel regridding collisions**
 - Shared weight files can race in multi-job environments
 - File locking and explicit cache hygiene matter
- **Lesson:** many production failures are technical wrappers around scientific assumptions

Scientific Lessons - What This Pipeline Really Teaches

- **Lesson 1:** preprocessing is part of the science
- **Lesson 2:** parity to a research-developed method is a major asset when moving toward operations
- **Lesson 3:** BC and ML answer different parts of the problem
- **Lesson 4:** maps, PDFs, VALUE, and residual views are all needed together
- **Lesson 5:** a climate-useful method must preserve the future signal, not only score well in the present
- **Operational corollary:** a cleaned workflow is what lets the science survive handoff, reproducibility, and production scaling

Summary And Takeaways

- **Key idea:** AI downscaling is useful when climate science, preprocessing, and ML design stay aligned
- **Reliability requires**
 - Stable datasets
 - Explicit scaling and calendar handling
 - Validation beyond one scalar score
- **Practical takeaway:** many of the worst failures happen before model depth even matters
- **Forward-looking takeaway:** once the baseline is trusted, then it becomes meaningful to optimize, extend to other variables, and prepare demonstrator production

Resources And Q&A

- **Repository:** github.com/cerfacs-globc/idownscale
- **Portable setup:** `environment.yml` and `doc/ENVIRONMENT_SETUP.md`
- **Workflow entry point:** `bin/production/run_exp5_workflow.py`
- **Collaborators:** CERFACS, University of Belgrade
- **Acknowledgements:** IRISCC Horizon Europe Project, Zoé Garcia



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- **Questions?**

Keywords: Climate AI, U-Net, Downscaling, Preprocessing Sensitivity, Numerical Stability, HPC Integrity

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What Archival Parity Means Here

- **Context:** the original workflow emerged from scientific exploration, then we refactored it into a cleaner and more flexible implementation
- **Goal of the restoration work:** recover the same scientific workflow and same end behavior, not quietly invent a new method
- **So parity was checked at several levels**
 - Preprocessing outputs
 - Packaged samples
 - Archival checkpoint replay
 - Final historical metrics
- **Why this matters:** once parity is back, we can discuss improvements from a trusted baseline instead of debugging in the fog

Historical Validation We Can Trust

- **Monthly metrics are extremely close to archive**
 - Spatial correlation and bias are nearly identical
 - Daily and monthly behavior both match the archival evaluation closely
- **Interpretation:** the restored end-to-end workflow is not just runnable; it is scientifically anchored to the historical method
- **Reading note:** the figures on this slide show regenerated outputs; the closeness to archive comes from the explicit metric comparisons performed alongside them
- **This gives us a trustworthy base for**
 - Short-course teaching
 - Demonstrator production planning
 - Later scientific improvements